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Novel ternary deep eutectic solvents pretreatment of corn stalk to realize high-value utilization

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ABSTRACT

The high-valued utilization of biomass components through pretreatment technology can effectively alleviate the pressure of energy and environment. In this study, a ternary deep eutectic solvent (TDES) was prepared by using glycolic acid (GA) and ethylene glycol (EG) as double hydrogen donors and benzyl trimethyl ammonium chloride (TMBAC) as hydrogen acceptor. The corn stalk was separated by TDES to obtain cellulose-rich solid residue and low-condensed lignin. Moreover, the solid residue was converted into bio-oil by rapid pyrolysis, and the low-condensed lignin was converted into fluorescent lignin by grafting luminol for information encryption. The results showed that under the pretreatment condition of 150℃ for 3 h, TDES with the molar ratio of 1:1:2 had a lignin removal ratio of 90.60 %, and the yield of levoglucosan obtained from solid residue pyrolysis bio-oil reached 36.38 %. Furthermore, TDES prevented lignin from forming a C-C polycondensation structure and endowed it with polyhydroxy properties, which was beneficial to graft luminol to enhance the fluorescence intensity efficiently. The high-intensity fluorescent grafted lignin was used for information encryption of various materials. The mechanism of TDES pretreatment and luminol grafting were explored, and the mass balance of corn stalk during pretreatment was described. This study provides a green, environmentally friendly, and efficient pretreatment process for the high-value utilization of biomass.

1. Introduction

Lignocellulose biomass was the only carbon-based substitute for traditional fossil energy [1], which can be transformed into high-value products in many ways to optimize the social economy and bring environmental benefits [2]. However, the stubborn structure of biomass inhibited the efficient transformation and utilization of its components [3]. Typically, the recalcitrance of biomass arises from the covalent cross-linking of lignin and carbohydrates through benzyl ethers and esters, as well as weaker interactions such as hydrogen bonds and van der Waals forces. These interactions contribute to the inherent rigidity and complexity of the biomass structure, making it challenging to efficiently convert and utilize its components [4]. Therefore, biomass pretreatment has become a crucial processing step in the conversion of valuable components within biomass.

Lignin was usually treated as waste in the traditional biomass pretreatment or papermaking pulping process [5], leading to not only a waste of resources but also environmental pollution. Recently, researchers realized that the high-value utilization of lignin was essential to the economic feasibility of the lignocellulose bio-refining industry [6–8]. The research goal was usually to obtain active lignin for catalytic depolymerization into aromatic monomers (such as phenol, guaiacol, and 4-ethyl toluene) [9] and prepare multi-purpose bio-based products (such as adhesives [10], antibacterial films [11], sunscreens [12], etc.). However, lignin was susceptible to condensation during the traditional biomass pretreatment process, leading to the formation of recalcitrant C-C bonds that limited its subsequent conversion and utilization [13]. Specifically, solvents not only cleave the weak C-O bonds prevalent in lignin but also protonate the α-OH groups present in various lignin structures under acidic catalysis [14]. This leads to the generation of benzyl carbocations, which participated in electrophilic substitution reactions with the electron-rich aromatic rings of lignin, ultimately producing more difficult-to-degrade, stubborn C-C bonds [6]. Therefore, it was necessary to prepare lignin with a structure analogous to natural

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lignin, characterized by abundant β-O-4 linkages and low condensed structures. In recent years, several pretreatment solvents such as alkaline solution, organic solvent [15], and ionic liquid [16] have been employed to effectively isolate high-quality lignin, but these methods have not fully addressed the aforementioned issues. Shuai et al. [14] used formaldehyde in the pretreatment process to block the reactive groups that led to the condensation of lignin C-C bonds, which made the depolymerization performance of lignin close to the theoretical monomer yield. Although a high yield of lignin monomer was obtained in this treatment method, it inevitably involved the use of toxic reagents, such as 1,4-dioxane and formaldehyde. Consequently, a requirement for efficient, environmentally benign, and low-toxicity pretreatment methodologies that facilitate the enrichment of cellulose components while concurrently yielding valuable low-condensed lignin for potential applications.

The high recyclability, low cost, and low toxicity of deep eutectic solvents (DES) demonstrated exceptional performance in biomass pretreatment [17]. Nevertheless, Duan et al. [18] indicated that lignin extracted by binary DES in acidic systems underwent severe condensation and manifested drawbacks such as high molecular weight and low reactivity, which were unfavorable for further applications of lignin. Ethylene glycol can effectively protect valuable aryl ether bonds (β-O-4) and prevent lignin condensation during DES pretreatment by stabilizing carbocations [19]. Consequently, the integration of ethylene glycol into binary acidic DES presents a feasible approach for simultaneously separating biomass components and rationally modulating the structure of lignin. Besides, previous studies [20] showed that glycolic acid, as a donor, can effectively esterify the γ position of the β-O-4 bond and provide –OH reaction site. Nathan et al. [21]confirmed that lignin can be synthesized into polymer chains by “graft copolymerization” to obtain the characteristics of graft reagent. Luminol was widely used in forensic science [22], environmental monitoring [23], medical diagnosis [24], and other fields due to its bright fluorescence performance. Therefore, it was extremely possible to obtain the fluorescence properties of luminol according to the characteristics of non-condensed lignin with polyhydroxy groups.

In this study, the ternary deep eutectic solvent (TDES) was prepared by using glycolic acid (GA) and ethylene glycol (EG) as double hydrogen donors and benzyl trimethyl ammonium chloride (TMBAC) as hydrogen bond acceptor, and corn stalk was pretreated to obtain cellulose-rich solid residues and low-condensed lignin. The solid residues were converted into bio-oil rich in levoglucosan through fast pyrolysis, while low-condensation lignin was modified into fluorescent grafted lignin by grafting luminol for information encryption. Furthermore, the mechanisms of TDES pretreatment and grafting reaction were thoroughly investigated. Consequently, this study presents an efficient, eco-friendly, and low-cost novel process for the high-value conversion of biomass.

2. Materials and methods

2.1. Materials

Corn stalks, pine, poplar, and bagasse were purchased from Lianyungang City, Jiangsu Province. The raw materials were crushed, dried and screened with 60–80 mesh before the experiment. Glycolic acid (98 %, GA), ethylene glycol (98 %, EG), benzyltrimethylammonium chloride (98 %, TMBAC), methanol (99.5 %), anhydrous ethanol (99.5 %), N,N-dimethylformamide (99.5 %, DMF), luminol (98 %, Lum), 4,4′-Diphenylmethane diisocyanate (98 %, 4,4′-MDI) and dibutyltin dilaurate (95 %, DBTD) were purchased from Shanghai McLean Biochemical Co., Ltd. All experiments used distilled water and the experimental results were repeated three times to take the average.

2.2. Preparation and characterization of TDES

Firstly, benzyltrimethylammonium chloride (TMBAC), glycolic acid (GA), and ethylene glycol (EG) were mixed into a 250 mL pressure bottle with three different molar ratios of 1: 1: 1, 1: 1: 2, and 1:1:3, respectively. Subsequently, the mixture was heated in an oil bath pot at 80℃ until it was clear and transparent to obtain ternary deep eutectic solvent (TDES). The Kamlet-Taft parameters of the TDES were determined based on the UV absorption peaks of three dyes. Specifically, N,N-diethyl-4-nitroaniline (NEt_2), p-nitroaniline (NH_2) and nile red (NR) dyes were used as chromotropic indicators to determine the α, β, and π* values of TDES. Firstly, the dye was dissolved in methanol at a concentration of 1.0×10^{-3} mol·L^{-1}. Then, 500 μL of the dye solution was placed in a centrifuge tube and dried in an oven at 60 °C. Afterward, 4 mL of the TDES was added to the centrifuge tube to mix with the dye, which was then transferred to a cuvette. The absorption spectra at room temperature were recorded using a UV–Visible spectrophotometer (UV755B, CHN). The values of α, β, and π* were calculated by Eq.(1):

$$\begin{aligned} \pi^* &= 0.314 \times (27.5 - \nu_{NEt2}) \\ \beta &= 11.314 - \frac{3580}{\lambda(NH_2)_{\max}} - 1.125\pi^* \\ \alpha &= \frac{19.9657 - 1.0241\pi^* - V_{NR}}{1.6078} \end{aligned} \tag{1}$$

Where λ_{max} is the wavelength corresponding to the maximum absorption of dye, and $\nu = 1 / (\lambda_{max} \times 10^{-4})$.

2.3. Component separation of corn stalk by TDES

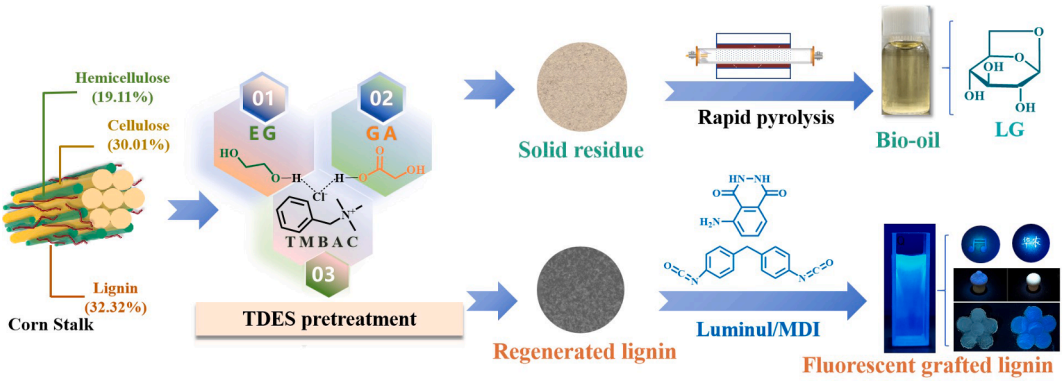
As shown in Scheme.1, the pretreatment experiments were conducted in a 300 mL pressure bottle. 3 g of corn stalk was mixed with 30 g of TDES and subjected to a pretreatment reaction at various temperature and time. After the reaction was completed, 20 mL of an acetone/water solution (volume ratio = 1:1) was added to quench the reaction. Then, the mixed solution was separated using a vacuum filter to obtain solid residues and liquid components. The solid residue was further washed with distilled water till pH = 7. Subsequently, the solid residue was dried in a vacuum oven at 80℃ for 24 h. Then, 300 mL distilled water was added to the liquid component containing TDES, and the lignin was precipitated after standing for 2 h. Subsequently, the precipitated solid was collected after repeated centrifugation and subjected to freeze-drying for 48 h to obtain the regenerated lignin. Finally, the excess water and acetone in the centrifuged liquid were removed by rotary evaporation to recover DES.

2.4. Preparation and application of fluorescent grafted lignin

In this synthesis process, 0.20 g of luminol (Lum) and 0.25 g of 4,4′-methylenebisphenylene diisocyanate (4,4′-MDI) were first dissolved individually in 10 mL of N,N-dimethylformamide (DMF) in two separate reaction flasks. The solutions were then stirred in an oil bath at 60℃ for 1 h in each flask. Then, the solution of luminol (Lum) in DMF was carefully dripped into the second reaction flask containing the MDI solution, and the mixture was agitated at a constant temperature for an additional 2 h. Subsequently, 0.1 g of regenerated lignin was added to the reaction flask, and 3–4 drops of DBTD (dibutyltin dilaurate) were introduced into the mixture as a catalyst. After the reaction, anhydrous ethanol was added to the reaction mixture to precipitate the product. The precipitate was filtered, rigorously washed multiple times, and then subjected to vacuum drying at 60 °C for 12 h, resulting in the synthesis of fluorescent grafted lignin, denoted as Lignin-Lum-MDI.

2.5. Pyrolysis of solid residue

The rapid pyrolysis experiment of solid residue was completed in a fixed-bed reactor. In short, 0.50 g of pretreated solid residue was placed in the center of a quartz tube (500 mm in length and 35 mm in



Scheme 1. Schematic illustration of TDES pretreatment process.

diameter). N_2 with a flow rate of 450 mL·min^{-1} was used to remove the air in the tube. The quartz tube with the sample was placed in the pyrolysis furnace and kept for 30 min when the temperature reached 500℃. During the whole process, N_2 was used as a carrier gas to carry the pyrolysis volatiles to the condenser at −18℃, and the bio-oil was collected with acetone. Finally, the components of bio-oil obtained after pyrolysis were determined by the GC–MS system (Agilent 7890B-5977A, USA). The relative content of bio-oil products was calculated by peak area ratio, and the absolute yield of levoglucosan was calculated by external standard method.

2.6. Characterization methods

2.6.1. Characterization of solid residue

The main component of cellulose, hemicellulose, and lignin was analyzed by the method in previous research [20]. The solid yield, cellulose retention ratio, and hemicellulose/lignin removal ratio were calculated by Eq.(2):

$$\text{Yield of solid residue (\%)} = \frac{\text{Solid component (g)}}{\text{Original biomass quality (g)}} \times$$

$$\text{Cellulose retention ratio (\%)} = \frac{\text{Cellulose relative content (g)}}{\text{Original cellulose content (g)}} \times \text{Yield of solid residue (\%)}$$

$$\text{Hemicellulose removal ratio (\%)} = 100\% - \frac{\text{Hemicellulose relative content (g)}}{\text{Original hemicellulose content (g)}} \times \text{Yield of solid residue (\%)}$$

$$\text{Lignin removal ratio (\%)} = 100\% - \frac{\text{Lignin relative content (g)}}{\text{Original lignin content (g)}} \times \text{yield of solid residue (\%)} \quad (2)$$

The crystallinity of solid residue was determined by a powder X-ray diffractometer (XRD, Rigaku Ultima IV, JPN). The test voltage and current were 40 kV and 40 mA, respectively. Additionally, the scanning angle range and speed of diffraction angle were $2\theta = 5 \sim 80$ and 8°·min^{-1}, respectively. The crystallinity index (CrI) and crystal size (D) were calculated by Eq.(3):

$$CrI(\%) = (I002 - Iam)/I002 \times 100\%$$

$$D = K\lambda/(\beta\cos\theta) \quad (3)$$

I_{002} and I_{am} meaned the peak intensity of 2θ around 22.5° and 18.5°, respectively. Additionally, λ was the radiation wavelength of the X-ray source (the value was 0.15406 nm), and β was the broadening of the diffraction peak. Moreover, K was the Scheler constant (the value is 0.9), and θ was the half diffraction angle. The TG/DTG curves of solid residue and regenerated lignin were tested by a thermogravimetric analyzer (TGA, Netzsch STA449C, GER). In details, 8–12 mg sample was placed in the Al_2O_3 crucible, and the temperature was raised from 30℃ to 800℃ at the rate of 15℃·min^{-1}, and high-purity nitrogen (30 mL·min^{-1}) was used as the carrier gas in the test process.

2.6.2. Characterization of regenerated lignin and fluorescent grafted lignin

The chemical structure changes of regenerated lignin and Lignin-Lum-MDI were analyzed by Fourier transform infrared spectrometer (FTIR, Brooker Vertex, GER). 2 mg sample and 100 mg KBr were accurately weighed and tabletted. The spectral analysis of the sample was carried out in transmission mode in the wavenumber range of 4000–400 cm^{-1}, and 32 scans were carried out with a resolution of 4 cm^{-1}. The regenerated lignin structure was analyzed by 2D-HSQC NMR (Bruker Avance Neo 600 MHz, GER). 50 mg lignin was completely dissolved in 0.5 mL DMSO-d_6 for testing and the reaction time was 10 h. The spectral widths of 1H and 13C dimensions were 1–15 ppm and 50–250 ppm, respectively. The data was processed by MestReNova software, and the related signals in the spectrogram were classified according to the published literature [25]. The integral of the $0.5S_{2,6} + G_2 + 0.5H_{2,6}$ signal was used as the internal standard, and the semi-quantitative analysis of lignin bond-related signals was calculated by Eq.(4):

$$IC_9 = 0.5S_{2,6} + G_2 + 0.5H_{2,6}$$

$$I_X\% = I_X/I_{C9} \times 100\ \% \quad (4)$$

I_A, I_B, and I_C represented the two-dimensional nuclear magnetic signal integrals of *β-O-4*, *β-β*, and *β-5* at α position, respectively. The solvent properties of TDES and aqueous solution of Lignin-Lum-MDI were analyzed by ultraviolet–visible spectrophotometer (UV755B, CHN). And the fluorescence intensity of the aqueous solution of Lignin-Lum-MDI was measured by a fluorescence spectrophotometer (RF-5301PC, JPN). The excitation wavelength was 365 nm, and the exit slit was 2.5 nm, and the PMT voltage was 700 V. The Gaussian 09 E.01 package was used for density functional theory (DFT) calculations in the experiment. Hybrid functional B3LYP and basis set 6-31G (d, p) were used to describe the geometric optimization of all atoms. The calculation of was realized energy gap value and charge differential density by using the combination of Multiwfn 3.8 and VMD 1.9.4 programs [26–28].

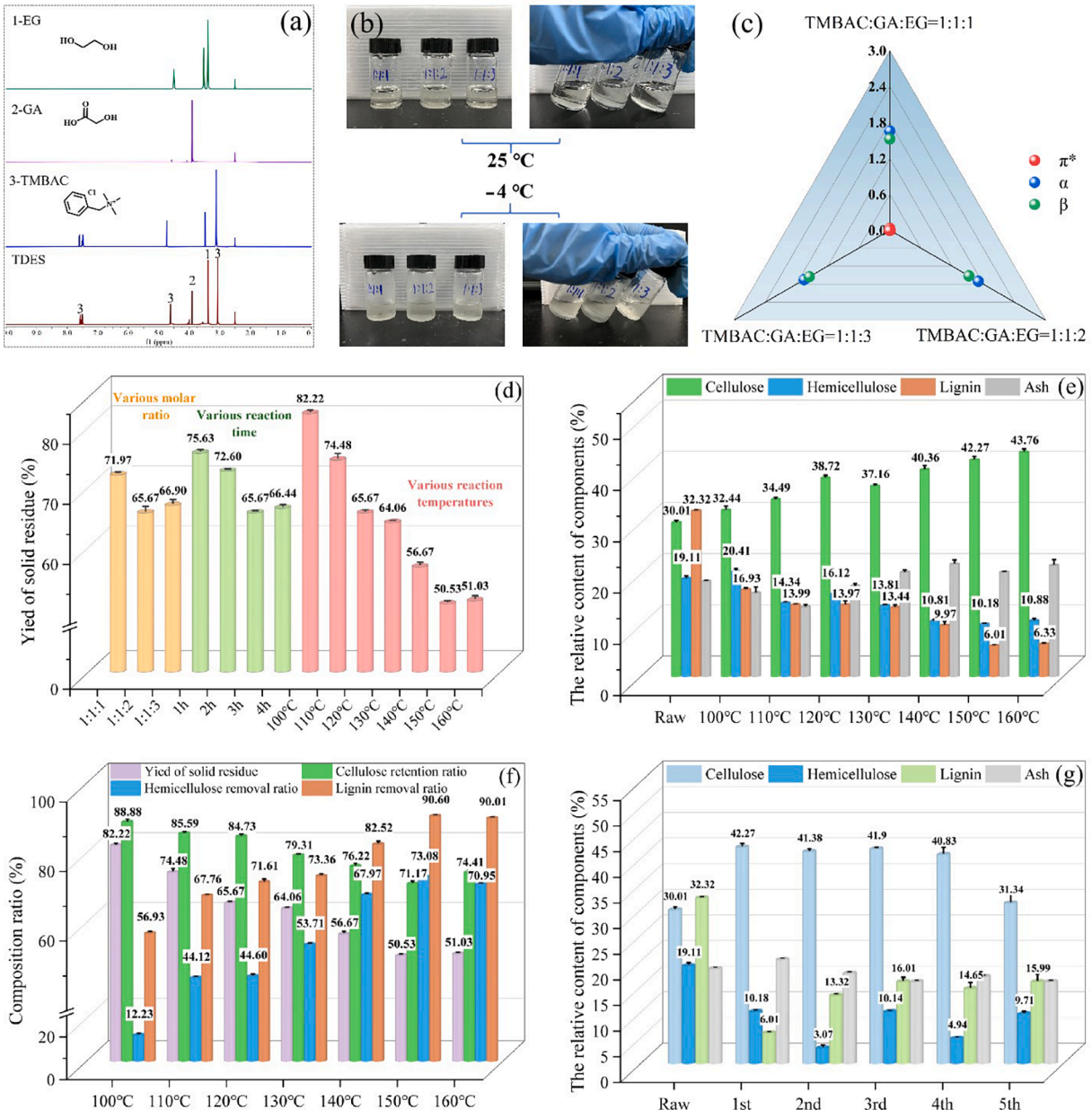


Fig. 1. (a) H^1 spectrum of TDES; (b) TDES prepared in different molar ratios; (c) K-T parameters of TDES;.(d) Yield of solid residue under different pretreatment conditions; (e) The relative contents of three components in the solid residue at different pretreated temperatures; (f) The change of composition of solid residue with pretreated temperature; (g) The cyclic pretreatment performance of TDES.

3. Results and discussion

3.1. Properties of TDES and efficiency evaluation of pretreatment

As shown in Fig. 1(a)-(b), the ternary deep eutectic solvent (TDES) was prepared with TMBAC, GA and EG. Firstly, the ^1H spectrum showed that the strong hydrogen signal peaks originally belonging to TMBAC, GA, and EG appeared in the TDES spectrum with slight peak shift, which was attributed to the formation of intermolecular hydrogen bonds among the three components [29]. Secondly, the prepared TDES remained in a liquid state at both 25 °C and −4°C, confirming the successful synthesis of a deep eutectic solvent. As shown in Fig. 1(c), the K-T empirical parameters were explored for different molar ratios of TDES. It was worth noting that the α values of TDES were 1.655, 1.707, and 1.652 with the increase of EG content, respectively. Previous studies have shown that TDES with a high α value can release more hydrogen protons and form a large number of hydrogen bonds to enhance the solubility of lignin [20]. Therefore, it was speculated that TDES at the molar ratio of 1:1:2 can remove more lignin and enrich cellulose.

The yield of solid residue of TDES pretreatment of corn stalks under different conditions was shown in Fig. 1(d). It was found that the yield of solid residue decreased from 71.97 % to 65.57 % with the increase of EG content in TDES, which may be attributed to two reasons. On the one hand, the increase of EG reduced the viscosity of TDES and promoted the penetration and destruction of biomass structure [30]. On the other hand, the cleavage of the C-O bond in lignin was accompanied by the formation of benzyl carbocation in the pretreatment of the acidic system, and EG can be grafted onto α-C to block the electrophilic aromatic substitution reaction among reactive intermediates and further inhibit lignin polycondensation [31]. Moreover, the yield of solid residue increased to 66.90 % when the molar ratio of TMBAC:GA:EG was 1:1:3, which indicated that the ability of TDES to break the glycosidic bond in

hemicellulose and the β-O-4 bond in lignin was weakened [20]. It was because the high content of EG led to the decrease of the α value of TDES, and then the ability to remove lignin was weakened. With the increase of pretreatment temperature from 100℃ to 160℃, the yield of solid residue decreased from 82.22 % to 51.03 % at different pretreatment temperatures. It was possible that high temperature promoted the decomposition of biomass structure.

As shown in Fig. 1(e) and (f), it was shown that the changes of three components in the solid residue at different temperatures. The untreated corn stalk (Raw) contained cellulose (30.01 %), lignin (32.32 %) and hemicellulose (19.11 %). It was found that the relative content of lignin gradually decreased to 6.01 % with the increase of pretreatment temperature to 150℃. It was due to the fact that the release of hydrogen protons from GA in TDES, which destroyed the glycosidic bond among carbohydrates and lignin, and then formed a new strong hydrogen bond to dissolve lignin. Moreover, it was also found that the lignin removal ratio gradually increased from 56.93 % to 90.60 %, which proved that the solubility of TDES to lignin was enhanced with the increase of temperature. Besides, the relative content of hemicellulose gradually decreased from 19.11 % to 10.88 %, which was attributed to the fact that temperature increased the hydrolysis ability of hydrogen protons to hemicellulose, which was also corresponding to the changing trend of hemicellulose removal ratio from 12.23 % to 70.95 %. In addition, it was found that the relative content of cellulose increased from 30.01 % to 43.76 %, which proved that TDES pretreatment realized the enrichment of cellulose. However, the retention ratio of cellulose decreased from 88.88 % to 74.41 % as the pretreatment temperature was elevated to 160 °C. This loss of cellulose can be attributed to its hydrolysis under acidic and high-temperature conditions.

Furthermore, pretreatment experiments conducted on three other biomass materials (pine, poplar, and bagasse) demonstrated the universal applicability of the TDES pretreatment method. As shown in Fig.S1, the TGA indicated that the biomass materials were enriched with relatively higher amounts of cellulose. Additionally, Fig.S1(c)-(d) revealed that more than 74 % lignin removal ratio and 70 % hemcellulose removal ratio were achieved for all tested materials, indicating the advantage of TDES in the component separation for different types of biomass. The pretreatment experiments with different molar ratios presented in Fig.S1(e)-(f) proved that the ratio of 1:1:2 was more effective in removing lignin and enriching cellulose. In addition, the recycling experiments of TDES were conducted under the conditions of a molar ratio of 1:1:2, at 150℃ for 3 h. Initially, the used TDES was recovered after the removal of water and acetone through rotary evaporation, and the pretreatment performance was presented in Fig. 1(g). The results demonstrated that after five cycles, TDES maintained its effective lignin removal ability.

3.2. *Characteristics of solid residue and efficiency of pyrolysis saccharification*

TGA was used to analyze the thermal decomposition behavior of corn stalks and solid residue at different temperatures. As shown in Fig.S2 (a)-(b), the carbohydrates (hemicellulose and cellulose) were decomposed in the temperature range of 180℃ to 400℃ heating resulting in the release a large number of volatiles [32]. The DTG curve showed that TDES removed a large amount of hemicellulose, resulting in a weight loss rate less than Raw in the temperature range of 140℃-275℃. In addition, the thermal decomposition reaction of cellulose occurred at about 350℃, and the peak intensity gradually increased with the increase of pretreatment temperature, indicating that it retained relatively more cellulose. The above results showed that TDES can effectively enrich cellulose and was beneficial to pyrolysis saccharification. The

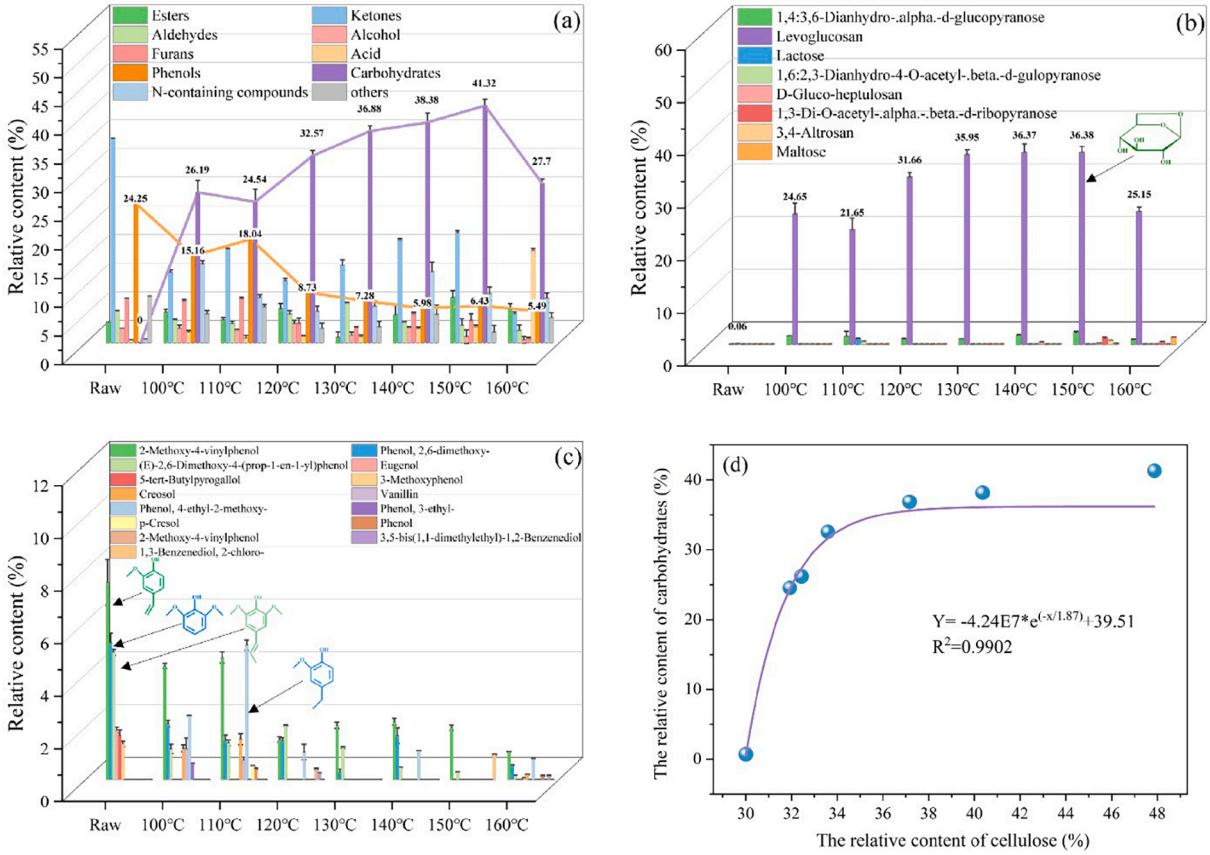


Fig. 2. (a) Product distribution of bio-oil; (b) The relative content of different carbohydrates in bio-oil; (c) The relative content of different phenols in bio-oil; (d) The correlation between relative content of carbohydrates in bio-oil and cellulose in solid residue.

crystallinity index (CrI) and crystal size (D) of residue were also important characteristics to evaluate the structural changes of corn stalk. As shown in Fig.S2(c)-(d), the typical diffraction peak intensities of crystalline cellulose I and lignin changed significantly. It was found that the CrI value increased significantly from 35.70 % to 53.77 % after TDES pretreatment, and the crystal size of CS decreased from 56.66 nm to 2.01–3.39 nm. It was because TDES pretreatment not only removed non-cellulose components (hemicellulose and lignin) but also reorganized and arranged amorphous regions to form a highly ordered crystal structure [33]. Specifically, TDES dissolved the lignin ring [34], and the amorphous region of corn stalk cellulose was invaded by hydrogen ions in GA, and the glycosidic bond was destroyed and a single crystal was released [35]. The high CrI value and low crystal size will increase the relative content of levoglucosan during pyrolysis.

The composition of residue pyrolysis bio-oil was shown in Fig. 2, which was mainly composed of carbohydrates, phenols, ketones, aldehydes, furans, nitrogen compounds, acids or ester products. During the pyrolysis, cellulose was converted into carbohydrates due to the breakage and dehydration of glycosidic bonds. However, hemicellulose was mainly decomposed into xylose after glycosidic bond cleavage, and then the intermediates were converted into hydroxypyranose, furfural and other oxygenates due to aldol conversion, ring opening and dehydration reactions [20]. The bio-oil of corn stalk without pretreatment was mainly composed of ketones (35.61 %) and phenols (24.25 %). With the increase of pretreatment temperature, the relative contents of ketones and phenols gradually decrease to 5.07 % and 5.49 %. This can be attributed to the fact that the increase in pretreatment temperature resulted in a relative decrease of the lignin content left in the solid residue. Furthermore, a substantial amount of β-O-4 and C-O linkages within the lignin cleavaged, leading to the production of phenolic, alcoholic, and ketonic compounds [36]. The relative content of aldehydes decreased from 5.54 % to 2.16 %, which indicated that it was related to the removal of hemicellulose. In addition, as the pretreatment temperature raised to 150 °C, the relative content of carbohydrates increased to 41.32 %. This result was attributed to two facts the enrichment of cellulose due to pretreatment. Moreover. the pores volume and specific surface area of residue gradually increased with the pretreated temperature, which promoted the outward diffusion of carbohydrate [37]. However, the relative content of carbohydrates decreased to 27.70 % when the pretreatment temperature increased to 160 °C due to the lignin condensation and crystallinity of cellulose reduction at high temperature [38]. As shown in Fig. 2(b) and (c), the specific components of carbohydrates and phenols were analyzed. The results indicated that levoglucosan was the dominant component among carbohydrates, achieving a maximum selectivity of nearly 95 %, which was attributed to the cleavage of glycosidic bonds and transglycosylation reactions within cellulose. Moreover, Fig. 2(d) showed that the content of carbohydrate in bio-oil was positively correlated with the content of cellulose in solid residue. The above results confirmed that TDES pretreatment effectively enriched levoglucosan in bio-oil by reducing lignin content and preserving a substantial amount of cellulose in corn stalk.

3.3. Structural analysis of low condensation regenerated lignin

It is well known that the polyhydroxy property of low-condensation

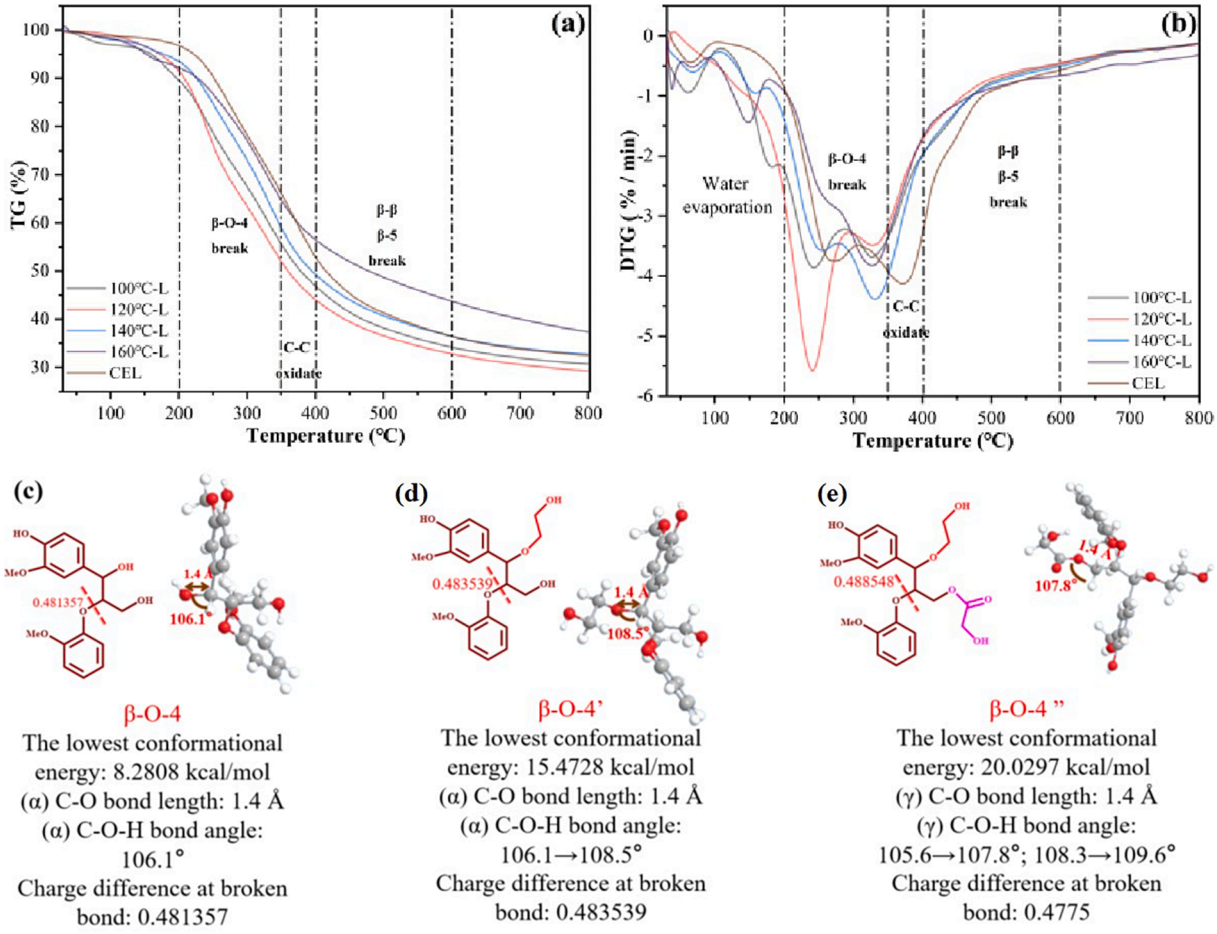


Fig. 3. (a)-(b) TG and DTG of regenerated lignin; (c)-(e) The lowest conformational energy, bond length and bond angle of lignin dimer model and the magnitude of charge at bond break.

lignin is beneficial to graft copolymerization [39]. As shown in Fig. 3 and Table S1, the TGA was employed to investigate the composition structure of regenerated lignin obtained under different pretreatment temperatures. It was obvious that the TDES had a profound impact on the structural properties of lignin. The depolymerization of lignin mainly involves chemical bond breakage (mainly β-O-4 and C-C in aryl ether structure) and side chain oxidation (carbonylation, carboxylation, dehydrogenation, etc.) [40]. Therefore, the weight loss of lignin during thermal decomposition positively correlates with the content of β-O-4 bonds [41]. Typically, the mass loss of lignin between 200℃ and 350℃ was mainly caused by the break of the β-O-4 bond. Under different temperature pretreatment conditions, the lignin extracted by TDES exhibited a mass loss of over 30 %, indicating that the regenerated lignin contained a significant amount of β-O-4 bonds. However, the β-O-4 bonds in lignin decrease with increasing temperature, possibly due to the condensation of lignin induced by high pretreatment temperatures [42]. In addition, the DTG curve also found that regenerated lignin contained two weight loss peaks in the range of 200-350℃. It was speculated that one was the β-O-4 bond retained in the original lignin, and the other was the shift peak of β-O-4 bond after grafting EG at α-C position [43]. Furthermore, the phenomenon was explained by constructing the model of lignin dimer and calculating the bond length, angle and charge difference of broken bond. As shown in Fig. 3(c)-(e), the conformations were adjusted by calculating the minimum energy conformations (MM2) of the molecules to achieve energy minimization. Fig. 3(d) showed that when EG was grafted at the α-position of the β-O-4 bond, the C-O bond length remained unchanged, but the C-O-H bond angle increased from 106.1° to 108.5°. The augmentation of the bond angle led to the creation of a larger space for electron mobility, which subsequently weakened the intermolecular bonding force, resulting in the bonds becoming more susceptible to breakage [44]. In addition, Fig. 3(e) revealed that the bond angle at the γ-position also enlarged when GA was grafted. Therefore, it was proved that the β-O-4′ bond connecting EG and GA was worse in thermal stability than the original β-O-4 bond in lignin. Moreover, the atomic charge magnitudes within the β-O-4 bond were computed, revealing that positions characterized by significant differences in charge magnitudes between adjacent atoms were comparatively more susceptible to fracture. In contrast, Fig. 3(c)-(e) showed that the grafting of EG increased the value of charge difference, but the difference decreased after grafting GA, indicating that the front peak in the DTG curve was mainly caused by the EG connection. The above results proved that TDES can effectively prevent the condensation of lignin and increase the relative content of β-O-4 bond in lignin [45]. By comparison, it was found that the 120℃-L contained more β-O-4 bonds and fewer β-β/β-5 bonds, which provided more grafted sites for various applications.

The chemical characteristics of regenerated lignin at different pretreatment temperatures were analyzed by FTIR. As shown in Fig. 4, the similar peak positions between the TDES-extracted lignin and the CEL spectrum indicate the structural integrity of the regenerated lignin. Generally, the broad diffraction peak near 3400 cm^{-1} corresponded to the –OH stretching vibration in lignin [16], and the absorption peaks from $–CH_2$ and $–CH_3$ appeared at about 2915 cm^{-1}. Besides, the absorption peak at 819 cm^{-1} corresponded to the C-H bending vibration out of the aromatic plane, which confirmed the existence of p-hydroxyphenyl unit (H-unit). Moreover, the absorption bands at 1602 cm^{-1}, 1459 cm^{-1}, and 1514 cm^{-1} have no significant variation with the aromatic skeleton corresponding to lignin [46]. In addition, the peaks at

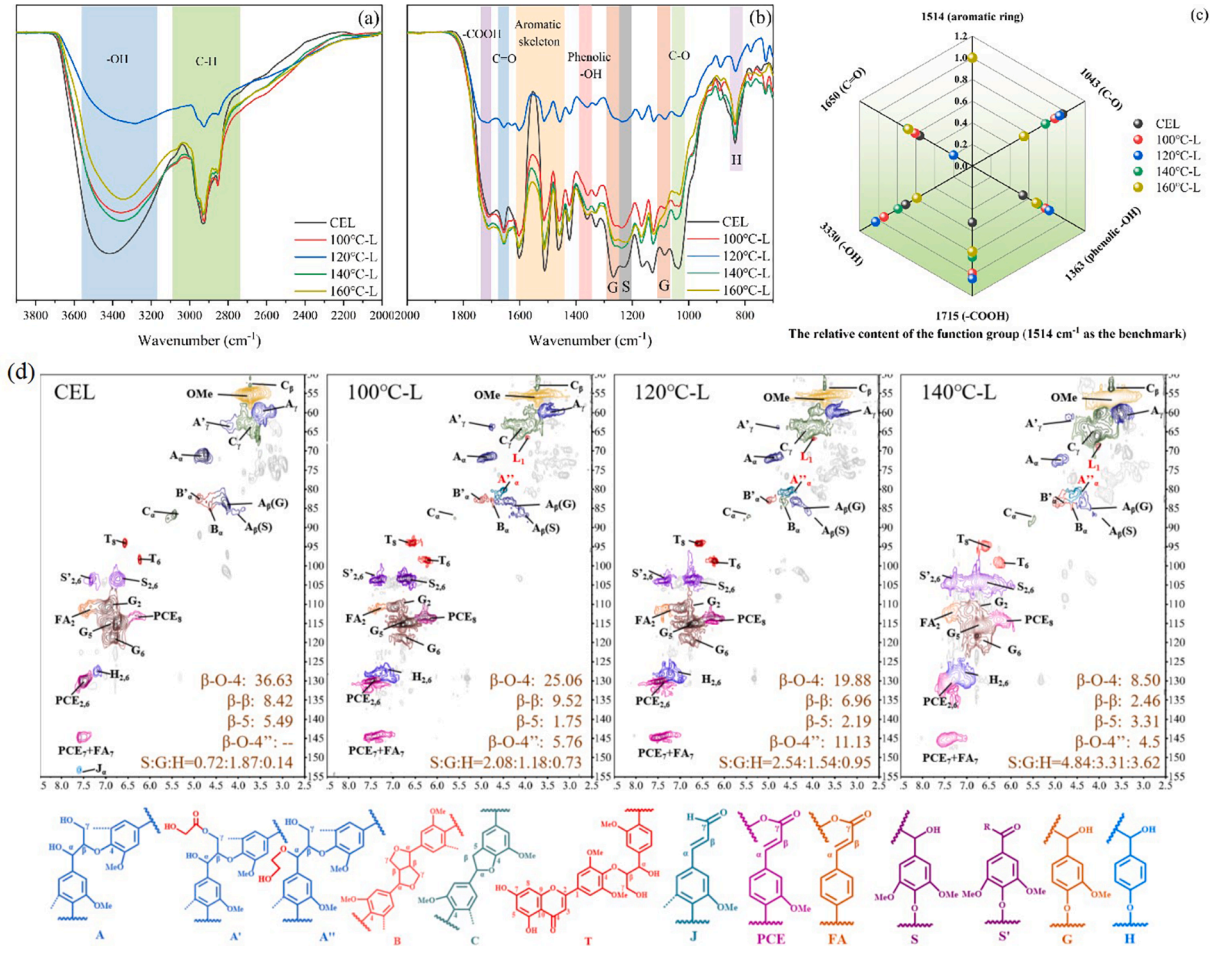


Fig. 4. (a) – (b) FTIR of regenerated lignin; (c) The relative content of function group of regenerated lignin; (d) 2D-HSQC NMR spectrum of regenerated lignin.

1148 cm^{-1}, 1032 cm^{-1}, and 1357 cm^{-1} were attributed to the characteristic signal of the guaiacol unit (G-unit), and the peak at 1255 cm^{-1} corresponded to the syringyl unit (S-unit) [20], which indicated that the structure of regenerated lignin did not disintegrate obviously during TDES pretreatment. Based on the changes in peak intensities of various functional groups in the FTIR, a semi-quantitative analysis was performed using the absorbance ratios of different functional groups as relative contents. The absorbance value of the aromatic ring (1514 cm-1) was set as a constant value of 1, and the calculation results are shown in Fig. 4(c). Generally, the characteristic peak (1650 cm^{-1}) of aromatic ketones mainly comes from C = O in the α position in Hibbert ketone structure, which is caused by the break of *β*-O-4 bond of lignin, which transforms the α-OH into C = O [47]. It was found that the C = O ratio of 120℃-L and 130℃-L were lower than CEL (0.56), which indicated that the *β*-O-4 bond breakage was inhibited, which was caused by EG grafted to lignin. However, the condensation of lignin made the content of C = O higher than CEL when the pretreatment temperature exceeded 130℃. At the same time, it was found that the content of –OH first increased and then decreased with the increase of pretreatment temperature, which also indicated that the introduction of EG formed α-etherified lignin which inhibited condensation, but the relative content was affected by temperature. In addition, the carboxyl content of regenerated lignin was higher than CEL, which was due to the connection of GA at γ position [20]. In general, TDES was grafted into the regenerated lignin structure to provide more –OH for downstream production.

The detailed structural information of regenerated lignin was analyzed by 2D HSQC-NMR, and the structural changes during pretreatment were observed with CEL as the control. The lignin cross-signal was referred in previous publications (Table S2) [25]. As shown in Fig. 4 (d), the spectrum of regenerated lignin can be divided into two regions: the side chain region (δC/δH 50.0–90.0/2.50–6.0 ppm) and aromatic region (δC/δH 95.0–155.0/9.0–5.5 ppm). In the side chain region, it was shown that the *β*-O-4 signal (Aα), *β*-*β* (Bα), and *β*-5 (Cα) of CEL obviously displayed at 71.62/4.85 ppm, 84.73/4.65 ppm, and 86.83/5.42 ppm for cross signals, and the relative contents were 36.63 %, 8.42 %, and 5.49 %, respectively by calculation. It was found that the signal intensity of α decreased with the pretreatment temperature from 100℃ to 140℃, and its relative content decreased from 25.06 % to 8.50 %, indicating that the original β-O-4 bond in lignin was gradually destroyed with the increase of pretreatment temperature. At the same time, a new signal of 80.05/4.51 ppm was found in the regenerated lignin spectrum, which was distributed to the structural derivative (A“α) of *β*-O-4, because EG was grafted to the α position of *β*-O-4 bond. In the pretreatment process of the acidic system, the *β*-O-4 bond of lignin was hydrolyzed to form highly reactive benzyl carbocation in the side chain of lignin, which attacked the electron-rich lignin aromatic rings and formed a stable C-C bond, leading to serious and irreversible polycondensation [14]. However, it was found that EG attacked the carbocation intermediate and formed a *β*-O-4″ structure which inhibited condensation in this study. Moreover, the relative content of the *β*-O-4″ bond increased from 5.76 % to 11.13 % and then decreased to 4.5 % with the increase of pretreatment temperature, which indicated that temperature had a strong influence on EG grafting. At the same time, the total content of *β*-O-4 bond and *β*-O-4″ bond in 120℃-L was 31.01 %, which was significantly close to CEL (36.63 %), indicating that 120℃ has the best lignin protection effect. Furthermore, when comparing the pretreatment efficacy of TDES on lignocellulosic biomass with prior research findings (Table 1), it was evident that the retention ratio of β-O-4 lignin in this study surpassed 84.65 %, confirming TDES's exceptional ability to preserve structural integrity. Moreover, the maximum lignin removal ratio achieved reached 90.01 %, underscoring TDES's outstanding component separation capabilities. In addition, the appearance of the L_1 signal at 63.0/4.20 ppm proved the successful esterification of GA at the γ position of the *β*-O-4 bond [20].

In the aromatic region, the signals of guaiacyl (G), syringyl (S), and p-hydroxyphenyl (H) units of lignin were identified in CEL. Besides, the signals of oxidized eugenol unit (S'), wheat flavone (T), *p*-coumaric acid (PCE), and ferulic acid (FA) were observed. After TDES pretreatment, it was found that the signals of $S_{2,6}$ and G_5 had no obvious signal offset with the increase in pretreatment temperature. This result may be due to the inhibition of the α-position condensation reaction of the *β*-O-4 structure by EG [53]. For more detailed information about the structural changes of lignin after DES pretreatment, S/G/H was calculated based on previous research methods [15]. Based on the CEL ratio of 0.72:1.87:0.14, it was found that the lignin extracted by TDES contained more S-units and H-units. Previous studies have shown that the existence of S-units was beneficial to form *β*-O-4 bonds, while G-units would promote the formation of *β*-5 bonds [13], indicating that lignin with low G-unit content extracted by TDES was less prone to polycondensation. In conclusion, the TDES pretreatment can not only protect lignin structure by grafting EG but also provide potential for lignin application by grafting GA.

The molecular frontier orbital energy and binding energy of the interaction interface were calculated by DFT to analyze the stability of the intermolecular ground state structure in the pretreatment process (the lignin dimer model was represented by two guaiacyl units connected by β-O-4 bond). The energy gap values ($\Delta E_U = E_{(LUMO,\ solvent)}$-$E_{(HOMO,\ dimer)}$) indicated the difficulty of a chemical reaction between two molecules. As shown in Fig.S4(a), the value of ΔE_U between the composition of TDES and Dimer were 0.1474, 0.2103, and 0.2486 a.u., respectively. The results showed that it was more feasible for TMBAC to provide π electrons to Dimer, which also showed that TMBAC was more conducive to weak interaction with lignin in the pretreatment process [3]. In addition, Fig.S4(b) showed that the binding energy of TMBAC and dimer was the largest (−13.6663 kcal/mol), indicating that it had the strongest hydrogen bond attraction in the pretreatment process [3] Moreover, it was found that the electron cloud in lignin generated by Cl^- was larger than other atoms in the electrostatic potential diagram of Dimer-TMBAC, suggesting that hydrogen bonding interaction occurred between Cl^- in TMBA and H bond in lignin.

Furthermore, a possible mechanism for the TDES pretreatment process was proposed based on the characterization results and DFT

Table 1
Comparison of several DES for lignocellulosic biomass pretreatment.

[1]DES system	Biomass	Pretreatment condition	[2] Protection ratio of β-O-4 bond (%)	Lignin removal ratio (%)	Ref.
ChCl:LA	Rice straw	130 °C-3 h	13.42 %	31.80 %	[48]
ChCl:EG	Bamboo	110 °C-1 h	15.83 %	69.72 %	[9]
ChCl:CA	Bamboo	120℃-1.5 h	70.70 %	78.81 %	[49]
ChCl:GA_2	Bagasse	100 °C-9 h	69.85 %	68.04 %	[50]
ChCl:LA-EtOH	Broussonetia papyrifera	140 °C-1 h	25.85 %	~80 %	[49]
ChCl:LA-Microwave	Corn Stalk	160 °C-0.5 h	14.70 %	89.75 %	[51]
ChCl:EG-NaAC	Corn Stalk	120℃–3h	72.91 %	88.10 %	[52]
TMBAC:GA:EG	Corn Stalk	120℃–3h	84.65 %	90.01 %	This work

[1] ChCl (Choline chloride), LA (lactic acid), EG (ethylene glycol), CA (Citric acid), GA (Glycolic acid), GA_2 (Glyoxylic acid), NaAC (Sodium acetate), TMBAC (Benzyltrimethylammonium chloride).

[2] Protection ratio of β-O-4 bond (%) = β-O-4 bond content of lignin extracted by DES / β-O-4 bond content of (MWL or CEL).

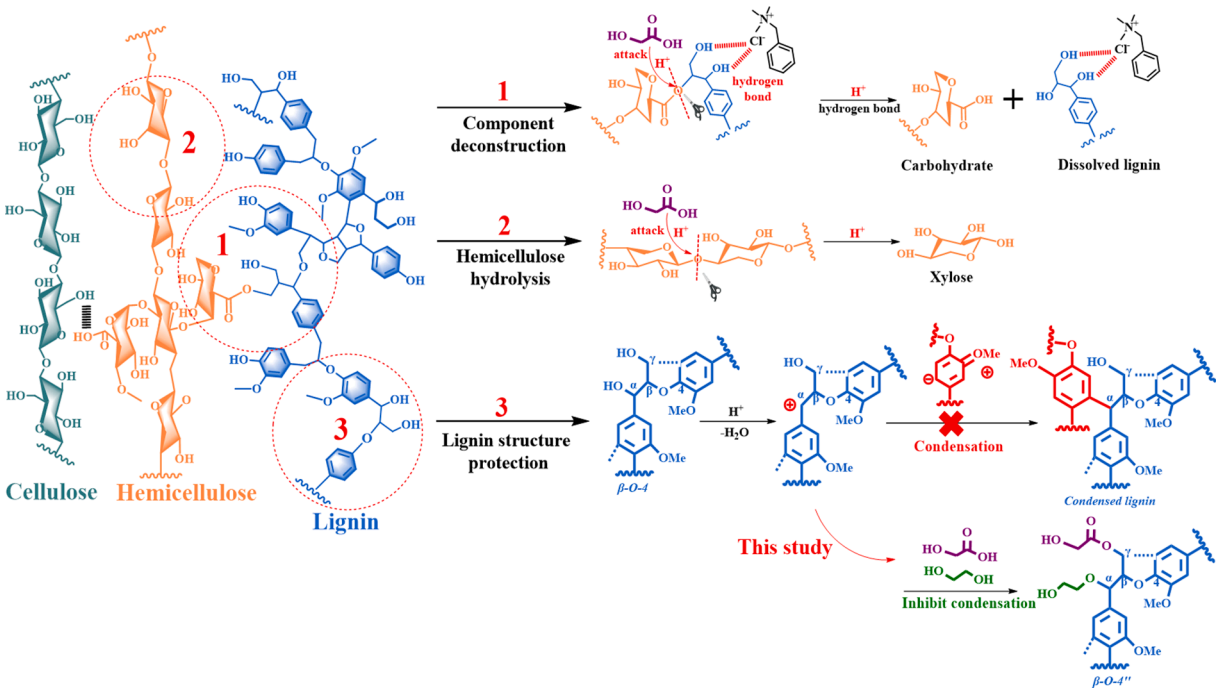


Fig. 5. The component separation mechanism of corn stalk by TDES.

calculation. As shown in Fig. 5, the process was roughly divided into three processes. Firstly, the hydrogen protons in GA destroyed the glycosidic bond between lignin and carbohydrates [20], while the Cl^- ions in TMBAC formed new hydrogen bonds with the hydroxyl groups of lignin [54], weakening the strong hydrogen bonding interactions between lignin and carbohydrates and thereby disrupting the recalcitrant structure of biomass. Secondly, the hydrogen proton released by GA destroyed the ether bond inside hemicellulose and promoted its hydrolysis to xylose [50]. Finally, the cross-linked network structure of lignin and carbohydrates (especially hemicellulose) was destroyed by hydrogen protons, and high-activity benzyl carbocation was formed with the side chain of lignin. Then, the benzyl carbocation attacked the electron-rich lignin aromatic ring to form a lignin-condensed structure containing a large amount of stubborn C-C bonds [55]. The condensation of lignin limited the subsequent value-added utilization of fractionated lignin. Therefore, the introduction of EG into TDES effectively quenched the benzyl carbocation intermediate at the α position of lignin and formed an ether bond, which effectively inhibited condensation of lignin [42]. At the same time, GA was effectively esterified at the γ position of the lignin *β*-O-4 bond, and the *β*-O-4″ bond was formed.

3.4. Grafted mechanism and application of fluorescent grafted lignin

The aqueous solution of Lignin-Lum-MDI with a concentration of 1 $mg{\cdot}mL^{-1}$ was prepared and applied, and the solution was analyzed by UV–vis spectra. As shown in Fig. 6(a), the peak at about 250 nm in UV–vis spectrum was attributed to π-π* transition in lignin conjugated carbon core [3]. Besides, the peak around 298 nm corresponds to the n-π* transition of −C = O group [56]. The weak absorption peak at about 350 nm produced by n-π* electron transition was related to the N content of Lum grafted with lignin [23]. It was observed that different pretreatment conditions altered the UV absorption intensity of lignin-Lum-MDI, which may be correlated with structural changes in lignin. Furthermore, Fig. 6(b) showed the fluorescence emission intensity of the Lignin-Lum-MDI aqueous solution under an excitation wavelength of 365 nm. It was observed that the emission wavelength was centered around 450 nm, indicating that the fluorescence emission was primarily blue light. It was also found that the fluorescence intensity initially increased and then decreased as the pretreatment temperature rose from 100 °C to 140 °C. Combined with nuclear magnetic resonance spectroscopy, it was observed that the total content of *β*-O-4 and *β*-O-4′ linkages increased from 30.82 % to 31.01 % and then decreased to 13.00 %, indicating that TDES inhibited the condensation of lignin, which facilitated the grafting of luminol and increased the fluorescence intensity. In addition, the chemical stability of Lignin-Lum-MDI extracted under different pretreatment conditions was analyzed by 3D fluorescence spectrum. As shown in Fig. 6(c)-(h), it showed that Lignin-Lum-MDI not only had dual excitation wavelengths (285 nm and 350 nm) but also kepted the emission peak position very well. Double excitation wavelength meaned that anti-counterfeiting fluorescent materials will show different fluorescence characteristics under the irradiation of two different wavelengths of light. The double verification mechanism not only greatly improved the security of anti-counterfeiting, but also had high stability [57]. Furthermore, it was demonstrated that TDES significantly enhanced the fluorescence intensity compared to binary DES pretreatment. This was due to the dual hydrogen donors of TDES effectively protecting the hydroxyl groups at the α and γ positions of the *β*-O-4 bonds in lignin, which subsequently enables the grafting of more fluorophores (luminol), thereby promoting intramolecular charge transfer [23].

Based on the protective mechanism of TDES on lignin, the fluorescent grafted lignin (Lignin-Lum-MDI) was prepared by grafting luminol, and the structural changes of lignin before and after grafting were analyzed by FTIR. As shown in Fig. 7(a), the FTIR spectrum of 4,4′-Diphenylmethane diisocyanate (MDI) revealed a broad peak at 2272 cm^{-1}, which corresponded to the −C = N group within the MDI molecule. Additionally, the two sharp peaks were present within the wavenumber range of 1500–1600 cm^{-1}, which were attributed to the −C = O and −C = C groups, respectively. In the spectrum of Lum, the strong $–NH_2$ stretching peak was revealed at 3200–3400 cm^{-1}. Besides, the range of –NH bending vibration peak was 1600–1660 cm^{-1}, and the appearance of peak at 1320 cm^{-1} was assigned to −C = N stretching. Compared with 120℃-L, the intensity of –OH tensile peak in the range of Lignin-Lum-MDI decreased and strong $–NH_2$ vibration from Lum appeared at 3000–3500 cm^{-1}. At the same time, the peaks of –NH and C = N attributed to Lum were reflected in 1652 cm^{-1} and 1303 cm^{-1}. It

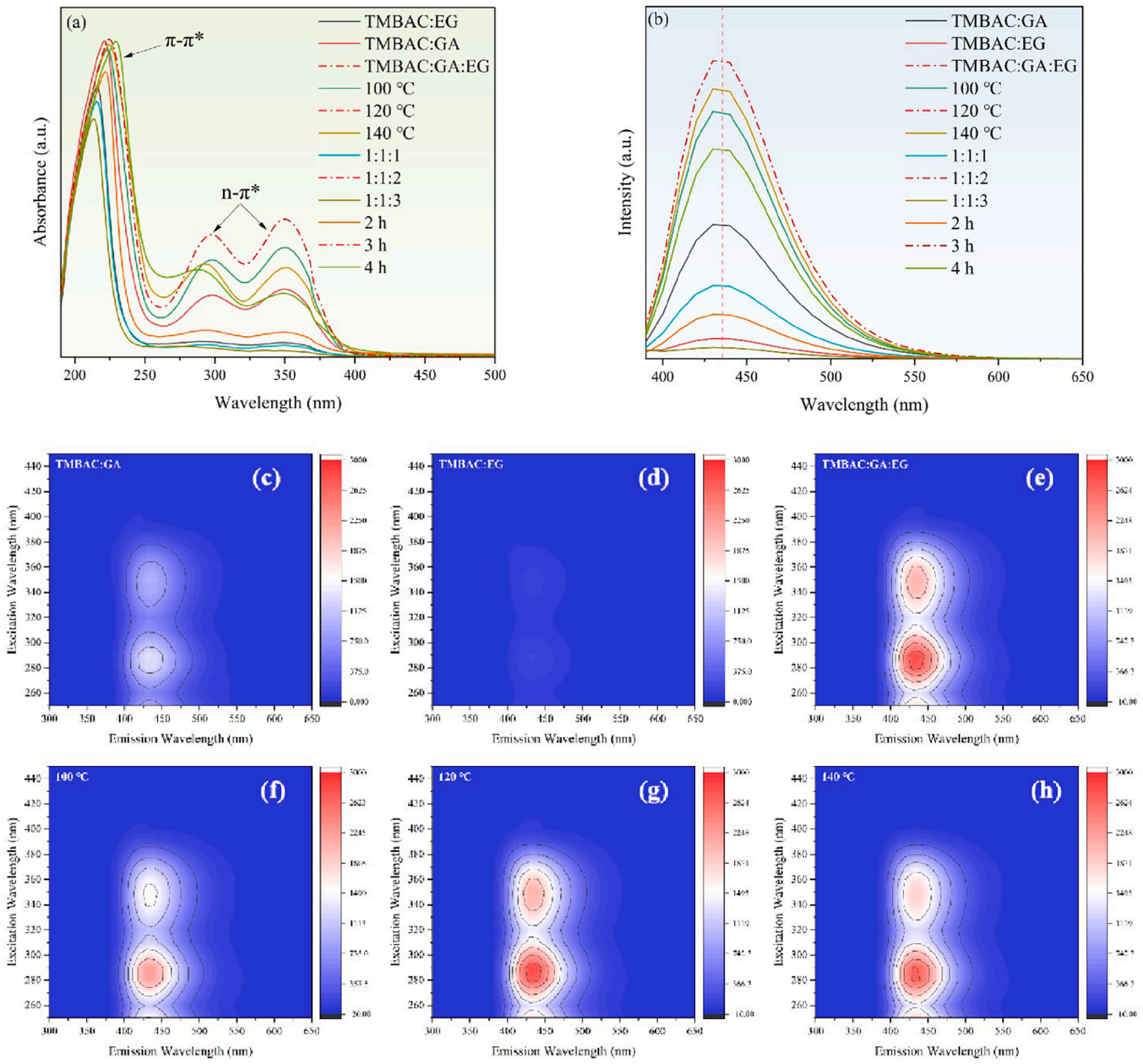


Fig. 6. (a) UV–vis spectra of the aqueous solution of Lignin-Lum-MDI; (b) Fluorescence emission intensity of the aqueous solution of Lignin-Lum-MDI was excited by 365 nm; (c)-(h) 3D fluorescence spectrogram of the aqueous solution of Lignin-Lum-MDI.

were clearly proved that Lum effectively grafted onto lignin by chemical crosslinking [23]. In addition, the appearance of 1600 cm^{-1}, 1510 cm^{-1} and 1425 cm^{-1} peaks of Lignin-Lum-MDI corresponded to the vibration band of aromatic skeleton, and the peak of transmittance at about 837 cm^{-1} was closely related to the C-H bending of p-hydroxyphenyl structure (H-unit) of lignin, which indicated that the phenylpropane structure of lignin was not damaged after grafting. In general, it was indicated that Lum was grafted on the –OH group of lignin. The $–NH_2$ groups present in Lignin-Lum-MDI were primarily attributed to the grafting of luminol. Therefore, the chemical structure of Lignin-Lum-MDI was analyzed by FTIR, and the content of $–NH_2$ was semi-quantitatively analyzed by the ratio of absorbance intensity of $–NH_2$ to the aromatic ring (1514 cm^{-1}) [3]. The high ratio exhibited by Lignin-Lum-MDI proved its high grafting efficiency. It was found that the lignin extracted under the reaction conditions of 1:1:2, 120℃, and 3 h had the highest grafting efficiency. At the same time, as shown in Fig. 7(b), semi-quantitative analysis of other functional groups revealed a positive correlation between the relative content of –OH in the regenerated lignin and the relative content of $–NH_2$ in the Lignin-Lum-MDI ($Y = 0.7X + 1.20$), indicating that regenerated lignin with a higher hydroxyl content favors the grafting of luminol. Additionally, Fig.S6(c) revealed that the Lum grafting efficiency followed a similar trend as the fluorescence intensity. Furthermore, a correlation analysis conducted in Fig. 7(c) demonstrated a linear positive relationship between the relative content of $–NH_2$ and the fluorescence intensity ($Y = 4527.53X-6182.63$), indicating that efficient Lum grafting contributed to the enhancement of fluorescence intensity. In addition, based on the above analysis, the mechanism of TDES pretreatment to enhance fluorescence intensity was proposed as shown in Fig. 7(d). Firstly, EG of TDES can effectively prevent lignin from forming a C-C condensation structure during pretreatment by quenching carbocation at the α-position of β-O-4 bond, and GA can be esterified at the γ-position, thus obtaining regenerated lignin with high –OH content. Subsequently, the –OH on the regenerated lignin was used as the grafting site for addition reaction with

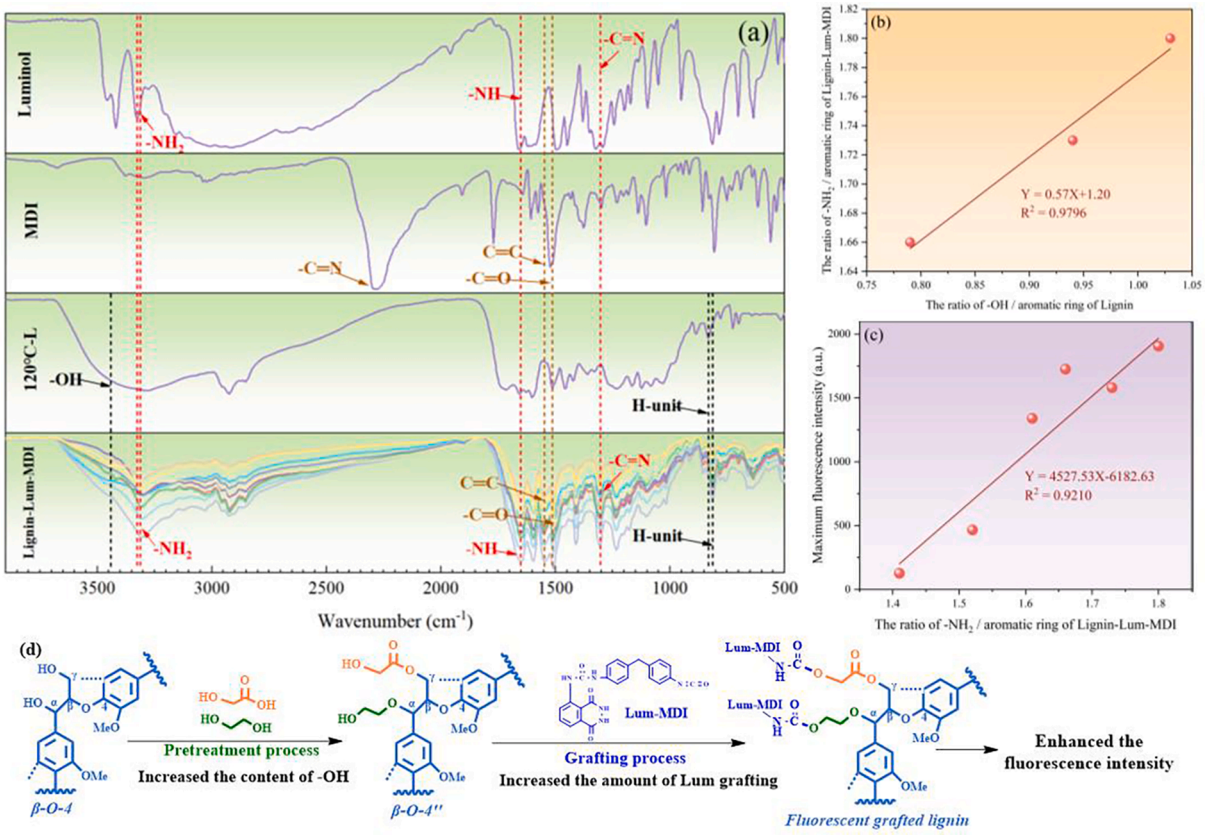


Fig. 7. (a) FTIR spectral of Lignin-Lum-MDI; (b) The correlation between the ratio of –OH/ aromatic ring of regenerated lignin and $–NH_2$ / aromatic ring of Lignin-Lum-MDI; (c) The correlation between the ratio of $–NH_2$ / aromatic ring of Lignin-Lum-MDI and maximum fluorescence intensity (365 nm excitation); (d) Schematic diagram of the grafting modification reaction process of lignin structure.

Lum-MDI to form fluorescent grafted lignin [23]. Therefore, TDES pretreated lignin increased the high grafting content of Lum-MDI and enhanced the fluorescence intensity of Lignin −Lum-MDI.

Charge density difference was usually used to compare the charge density before and after bonding to analyze the charge movement and polarization direction during the formation of chemical bonds [58]. As shown in Fig. 8(a)-(c), three kinds of lignin dimers were constructed to analyze the charge transfer during the grafting process of lignin and luminol. It showed that the electron density of –OH in Dimer decreased significantly, which was because the electron density distribution repels each other due to Pauli repulsion [59]. Besides, the density change between O-H had a cross section, which also reflected that the electron sharing degree of O-H bond was weakened. At the same time, a similar phenomenon also existed between C-N of Lum-MDI. Furthermore, the increased in charge density between the carbon (C) of Lum-MDI and the oxygen (O) of Dimer indicated that the grafting copolymerization primarily occurred through the addition reaction of the hydroxyl group (–OH) on Dimer with the isocyanate group (−N = C = O) on Lum-MDI. Therefore, it also proved that polyhydroxy lignin was beneficial to graft luminol efficiently. In addition, the distribution of electrostatic surface potential (ESP) of Lignin-Lum-MDI was calculated by DFT, and the minimum extreme value of EPS was calculated. As shown in Fig.S9 and Table S3, it was found that the absolute values of the most negative values at the O atoms of the side chain hydroxyl groups were all greater than those of the phenolic hydroxyl groups. Furthermore, the grafting reaction process of Lum primarily involved a nucleophilic addition reaction between the isocyanate (R-N = C = O) and the hydroxyl group (–OH), resulting in the formation of carbamate (−N-COO-R) [23]. In the nucleophilic reaction, the surface electrons of the O atom attack the N = C bond, thus a higher $V_{S,min}$ value of the O atom were beneficail to nucleophilic attack [60]. This indicated that the grafting of lignin and Lum-MDI preferentially occured at the β-O-4 hydroxyl positions of lignin.

As shown in Fig. 8(d), the difficulty of the reaction was analyzed by combining the energy profile diagram between Dimer and Lum-MDI [58]. It was found that Dimer-1 and Dimer-2 had the same energy of coordination (−8.0 kcal/mol), but the energy of transition state was reduced from 28.1 kcal/mol to 22.6 kcal/mol when EG was grafted at α-position. It proved that EG of TDES can effectively reduce the energy barrier between lignin and Lum-MDI to promote the grafting reaction. In addition, it was found that Dimer-1 and Dimer-3 had similar reaction energy barrier values, but Dimer-3 had lower product state energy, which indicated that the graft formed by Dimer-3 had a more stable chemical state. The above results showed that the lignin extracted by TDES not only endowed the lignin with high hydroxyl group characteristics, but also reduced the reaction energy barrier.

The Lignin-Lum-MDI prepared in this study had excellent fluorescence properties. The Fig. 9(a) showed pictures of aqueous solutions of different grafts under the irradiation of 365 nm ultraviolet excitation wavelength. It was found that H_2O, Lum, and 120℃-Lum-MDI were colorless and transparent under visible light, and showed dark blue when irradiated by UV light, but the aqueous solution of Lignin-Lum-MDI showed bright water-blue fluorescence, which was attributed to the charge transfer process in the molecule [23]. Previous studies have shown that the introduction of electron-withdrawing groups into fluorescent materials can greatly reduce the HOMO-LUMO gap, which leads to the change of fluorescence wavelength [56]. Therefore, the change of energy gap value (ΔE_U) of Lum before and after grafting lignin was calculated by DFT, as shown in Fig.S7. It was found that the ΔE_U of Lum was 0.15407, but it was reduced to 0.14742 after being grafted with lignin, and the electrostatic potential surface charge of luminol moved to the edge. On the one hand, it was proved that the introduction of lignin

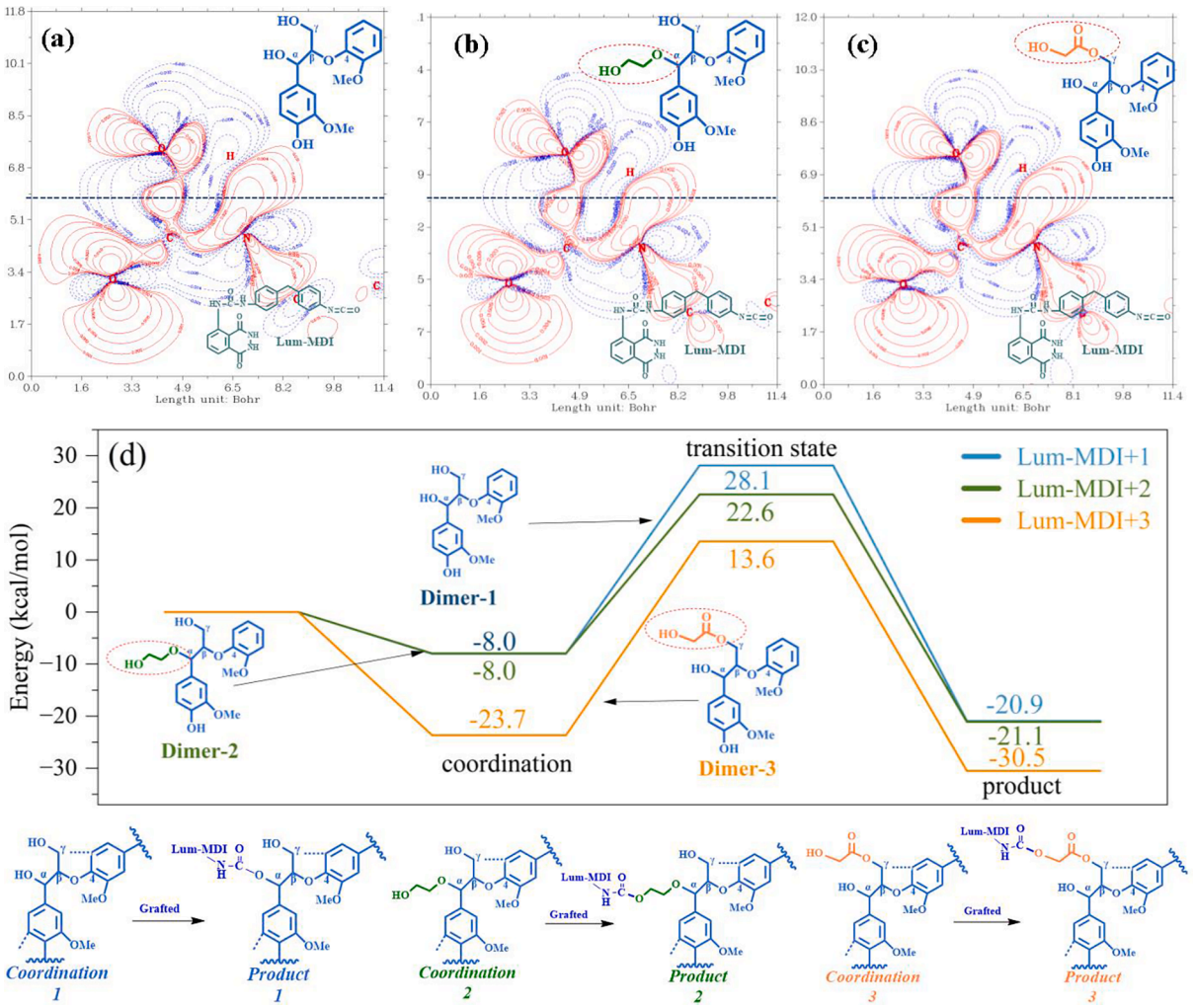


Fig. 8. (a)-(c) Charge density difference between Lum-MDI and Dimer-1, Dimer-2, and Dimer-3, respectively (Red represents the area with increased charge, and blue represents the area with decreased charge); (d) Energy profile diagram of the reaction process between Dimer and Lum-MDI.

led to a narrower band gap, which was conducive to reducing the energy gap. On the other hand, the electron cloud of Lignin-Lum-MDI was mainly distributed on Lum, which also proved that the lignin was the key factor to enhance fluorescence intensity.

Moreover, the Lignin-Lum-MDI at different pretreatment temperatures had a change in intensity visible to the naked eye, which was consistent with the result in Fig. 6. The anti-counterfeiting ink based on the bright fluorescence intensity of 120℃-Lum-MDI was applied to information encryption, as shown in Fig. 9(b). A variety of safety patterns (such as mice, pigs, signal peaks, musical symbols, leaves and small fish, etc.) were drawn on the filter paper, and the patterns appeared when exposed to ultraviolet light. Besides, the sample can also achieve bright fluorescence consistent with the anti-counterfeiting strip on the surface of the bank card. At the same time, the samples were also applied in wood products and gel products, which also showed excellent performance. Therefore, the Lignin-Lum-MDI developed in this study had the characteristics of low cost and strong fluorescence, and can be used for information encryption of various materials.

3.5. Mass balance of CS in pretreatment process

As shown in Fig. 10, the mass balance process of corn stalk during TDES pretreatment (based on pretreatment conditions of 1:1:2, 150℃ and 3 h) was demonstrated. The corn stalk (100 g) before pretreatment was composed of cellulose (30.01 g), lignin (32.32 g), hemicellulose (19.11 g) and ash (18.56 g). First, 50.53 g of solid residue mainly containing cellulose was produced after pretreatment with TDES. Then, it obtained 14.47 g of biochar, 10.69 g of synthesis gas and 25.08 g of bio-oil (containing 2.38 g of levoglucosan) from the residue by rapid pyrolysis. In addition, 49.47 g of components flowed into the liquid part (hemicellulose and lignin were the main components) after pretreatment. Then, 39.00 g grafted lignin was obtained after lignin regeneration and luminol grafting. After that, more than 80 % of TDES can be recovered by rotary evaporation to realize the process of recycling TDES. Generally speaking, the pretreatment method proposed in this study has high-efficiency pretreatment performance and high-value product application, which provides an efficient pretreatment new process for biomass refining.

4. Conclusion

In this study, a novel TDES was prepared by using GA and EG as double hydrogen donors, and the components of corn stalk were separated efficiently to prepare the high-value products of levoglucosan and fluorescent grafted lignin. TDES with a molar ratio of 1:1:2 was pretreated at 150℃ for 3 h, and the lignin removal ratio was 90.60 % and the hemicellulose removal ratio was 73.08 %, which realized the effective enrichment of cellulose. And the relative content of levoglucosan produced by pyrolysis and saccharification of residue reached 36.38 %. TDES still maintained excellent pretreatment ability after five

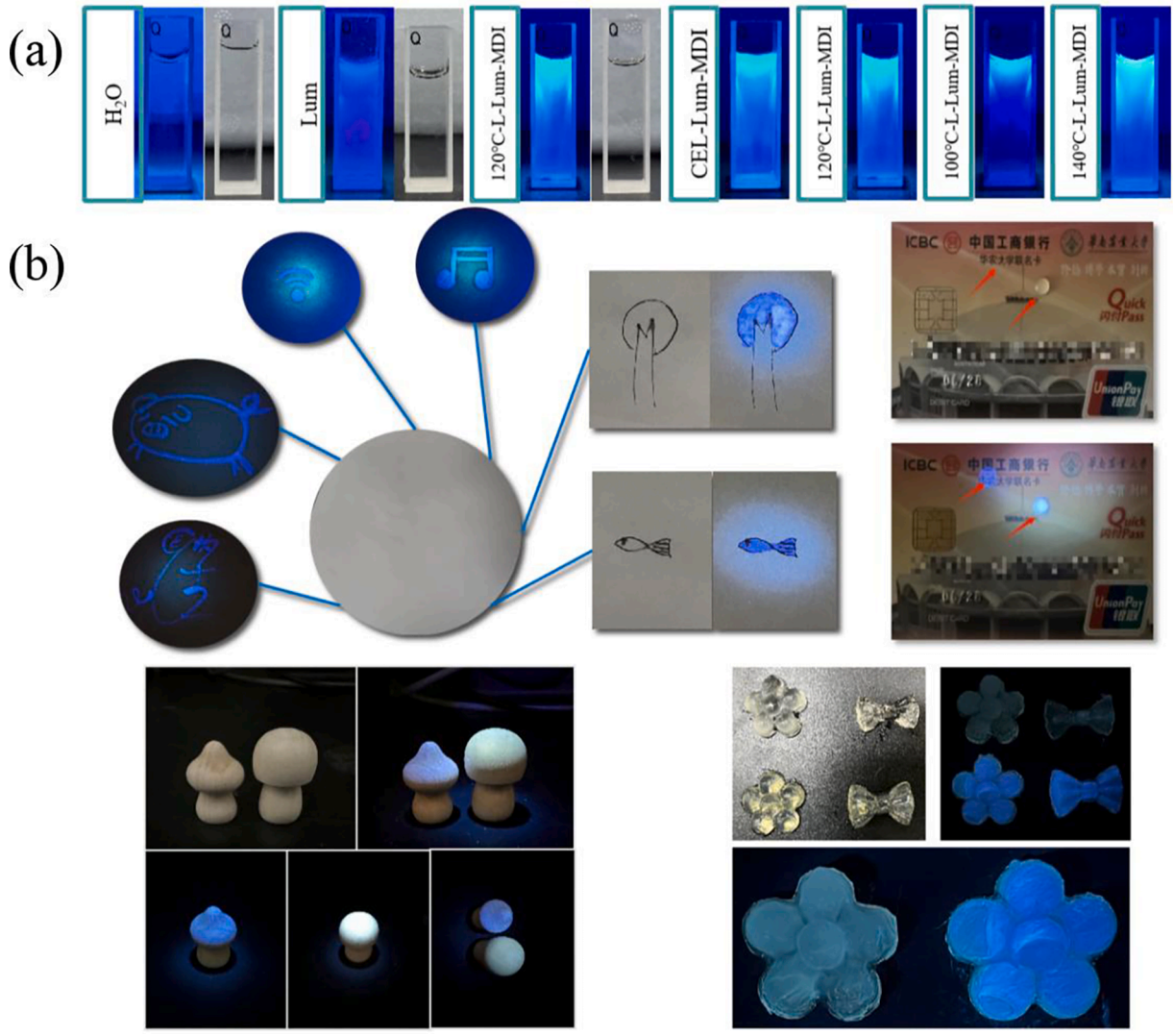


Fig. 9. (a) Photograph of fluorescent aqueous solution of Lignin-Lum-MDI; (b) Information encryption application (for paper, plastic cards, wood, hydrogels).

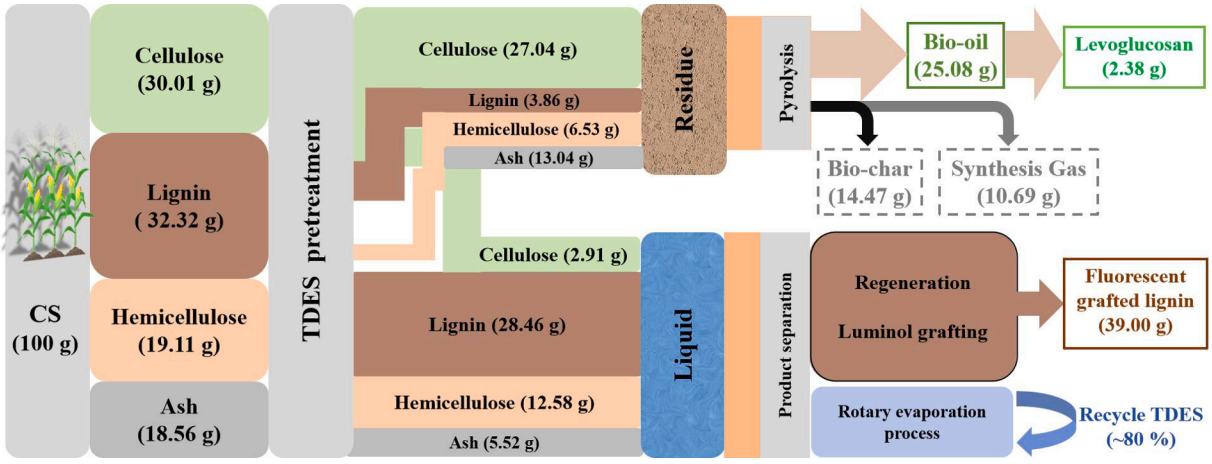


Fig. 10. The mass balance of corn stalk during TDES pretreatment.

cycles. In addition, TDES prevented the condensation reaction of lignin, and the content of β-O-4 bond in regenerated lignin reached 31.01 %. Moreover, the grafting based on double hydrogen donors endowed the regenerated lignin with polyhydroxy characteristics, thus achieving efficient grafting of Lum, which significantly promoted the increase of its fluorescence intensity. Fluorescent grafted lignin was widely used in information encryption of various materials. The mass balance also showed that the TDES prepared in this study had excellent separation

ability and recycling performance. The green and efficient pretreatment process in this study will inevitably bring rich environmental benefits and broad prospects for biomass refining.

CRediT authorship contribution statement

Zhiwen Jia: Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Xinlong Li:** Visualization, Validation, Data curation. **Shibo Xu:** Visualization, Validation, Data curation. **Yan Sun:** Writing – review & editing. **Lin Chen:** Visualization, Validation, Data curation. **Jiamin Ye:** Formal analysis, Data curation. **Wanyu Li:** Formal analysis, Data curation. **Tao Wu:** Formal analysis, Data curation. **Junyu Li:** Formal analysis, Data curation. **Shiyang Wang:** Data curation, Software. **Xiwei Xu:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2024.156567.

Data availability

No data was used for the research described in the article.

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